Problem set 3 – Econ 5030

1. Basic Cournot duopoly

Let begin by considering a Cournot duopoly. Each firm i has the cost function $C(q_i) = cq_i^2/2$, where $c > 0$. Market inverse $P(Q) = 200 - Q$.

1.a *Write the profit of the arbitrary firm i.*

$\pi_i = (200 - q_i - q_j)q_i - \frac{c}{2}q_i^2$.

1.b *Find the best response function for firm i and graph it on the $q_i \times q_j$ plane.*

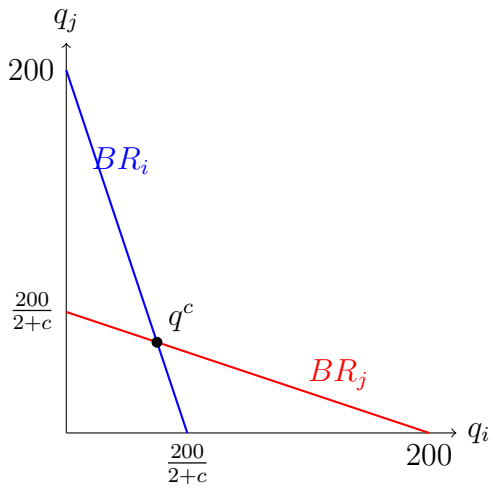
FOC $200 - 2q_i - q_j - cq_i = 0 \Rightarrow BR_i(q_j) = \dfrac{200 - q_j}{2 + c}$. Line from $(\frac{200}{2+c}, 0)$ to $(0, 200)$ (graph below).

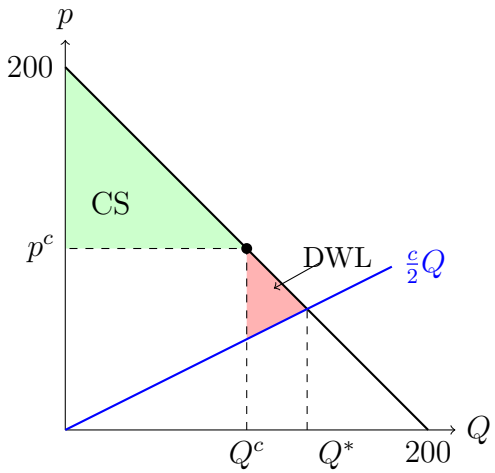
1.c *Find the Cournot equilibrium quantity q^c and market price $p^c = P(2q^c)$.*

Symmetric: $q^c = \dfrac{200}{3 + c}$, $p^c = \dfrac{200(1 + c)}{3 + c}$.

1.d *Graph the market outcome on the $p \times Q$ plane.*

Supply $= \sum MC$: $p = \frac{c}{2}Q$. Efficient $Q^* = \frac{400}{2+c}$, $p^* = \frac{200c}{2+c}$; Cournot at (Q^c, p^c), $Q^c < Q^*$.





1.e *Find an expression for the each firm's profits, the Consumer Surplus (CS), Total Surplus (TS) and Dead weight Loss (DWL).*

$\pi^c = q^c(p^c - \frac{c}{2}q^c) = \dfrac{20000(2 + c)}{(3 + c)^2}$, $CS = \frac{1}{2}(Q^c)^2 = \dfrac{80000}{(3 + c)^2}$, $TS = CS + 2\pi^c = \dfrac{40000(4 + c)}{(3 + c)^2}$.

$TS^* = \frac{1}{2} \cdot 200 \cdot Q^* = \dfrac{40000}{2 + c}$, $DWL = TS^* - TS = \dfrac{40000}{(2 + c)(3 + c)^2}$.

2. Basic Cournot oligopoly

Let now consider a Cournot oligopoly with n firms. Each firm i has the cost function $C(q_i) = cq_i^2/2$, where $c > 0$. Market inverse $P(Q) = 200 - Q$.

2.a *Write the profit of the arbitrary firm i.*

$\pi_i = (200 - q_i - Q_{-i})q_i - \frac{c}{2}q_i^2$.

2.b *Find the best response function for firm i and graph it on the $q_i \times Q_{-i}$ plane.*

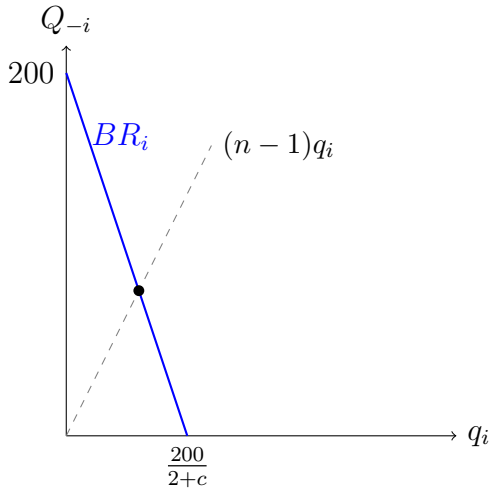
$BR_i(Q_{-i}) = \dfrac{200 - Q_{-i}}{2 + c}$. Line from $(\frac{200}{2+c}, 0)$ to $(0, 200)$; equilibrium where it meets $Q_{-i} = (n-1)q_i$ (graph below).

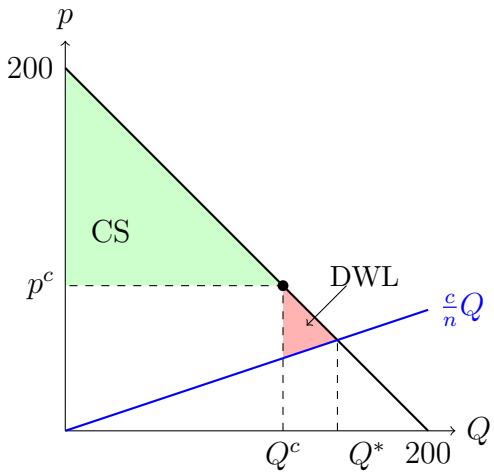
2.c *Find the Cournot equilibrium quantity q^c and market price $p^c = P(nq^c)$.*

$(2 + c)q^c = 200 - (n - 1)q^c \Rightarrow q^c = \dfrac{200}{n + 1 + c}$, $p^c = \dfrac{200(1 + c)}{n + 1 + c}$.

2.d *Graph the market outcome on the $p \times Q$ plane.*

Supply $p = \frac{c}{n}Q$; $Q^* = \frac{200n}{n+c}$; Cournot at (Q^c, p^c) with $Q^c = \frac{200n}{n+1+c} < Q^*$.





2.e *Find an expression for the each firm's profits, the Consumer Surplus (CS), Total Surplus (TS) and Dead weight Loss (DWL).*

$\pi^c = \dfrac{20000(2 + c)}{(n + 1 + c)^2}$, $CS = \dfrac{20000n^2}{(n + 1 + c)^2}$, $TS = \dfrac{20000n(n + 2 + c)}{(n + 1 + c)^2}$.

$TS^* = \dfrac{20000n}{n + c}$, $DWL = \dfrac{20000n}{(n + c)(n + 1 + c)^2} \to 0$ as $n \to \infty$.

3. Cournot with Product Differentiation

Take the inverse demand system: $P_1(q_1, q_2) = 18 - q_1 - \gamma q_2$, $P_2(q_2, q_1) = 18 - q_2 - \gamma q_1$. Assume both firms have NO COST of production. Now let us look at the Cournot solution for general γ between zero an one.

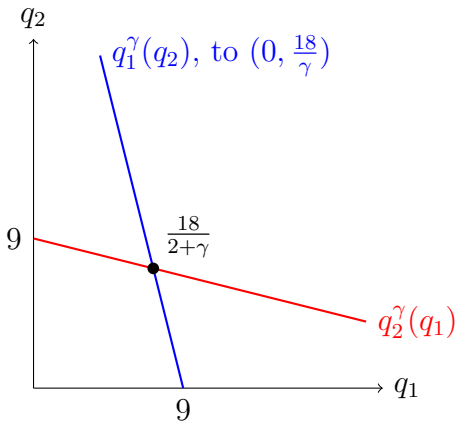
2.a *Write out each firm's profit function and label them $\Pi_1(\gamma)$ and $\Pi_2(\gamma)$.*

$\Pi_1(\gamma) = (18 - q_1 - \gamma q_2)q_1$, $\Pi_2(\gamma) = (18 - q_2 - \gamma q_1)q_2$.

2.b *Find each firm's best response function by solving $\frac{\partial \Pi_1(\gamma)}{\partial q_1} = 0$ for $q_1^\gamma(q_2)$ and $\frac{\partial \Pi_2(\gamma_2, \gamma_1)}{\partial q_2} = 0$ for $q_2^\gamma(q_1)$.*

$18 - 2q_1 - \gamma q_2 = 0 \Rightarrow q_1^\gamma(q_2) = \dfrac{18 - \gamma q_2}{2}$; $q_2^\gamma(q_1) = \dfrac{18 - \gamma q_1}{2}$.

2.c *Graph the best response functions of the firms' on $q_1 \times q_2$ space.*



2.d *Find the equilibrium quantities and prices and label them $q_1(\gamma)$, $p_1(\gamma)$ and $q_2(\gamma)$, $p_2(\gamma)$.*

$q_1(\gamma) = q_2(\gamma) = \dfrac{18}{2+\gamma}$, $\quad p_1(\gamma) = p_2(\gamma) = \dfrac{18}{2+\gamma}$ $\quad (\Pi_i = \frac{324}{(2+\gamma)^2})$.

4. Price competition

Suppose that thre are 2 firms in a Bertrand oligopoly with homogenous products. Each firm i has th cost of production $c(q_i) = q_i^2/2$. The industry demand function is the rectangle $D(p) = 0$ if $p > 10$, $D(p) = 10$ if $p \leq 10$. The if firms price the same then demand is split evenly. Each firm i has th cost of production $c(q_i) = q_i^2/2$. Based on this cost structure a firm will never supply more than their supply function $s(p_i) = p_i$. using this fact we can construct residual demand and firm profit. Firm profit is the piecewise function for prices in the interval $[0, 10]$ (I have done all the work for you here):

$$\pi_i(p_i, p_j) = \begin{cases} p_i \min\{p_i, 10 - p_j\} - \big(\min\{p_i, 10 - p_j\}\big)^2/2 & p_i > p_j \\ p_i 5 - 12.5 & p_i = p_j > 5 \\ p_i^2/2 & p_i = p_j \leq 5 \\ p_i^2/2 & p_i < p_j \end{cases}$$

1. *Because for most prices p_j the best response is not well-defined, instead find the argument of the supremum of the profit responding to the other firm's price. Formally find $\phi(p_j) = \arg\sup_{p_i \in [0,10]} \pi_i(p_i, p_j)$.*

Undercut: $\sup = p_j^2/2$ (not attained). Match: $\leq p_j^2/2$ (for $p_j > 5$, $p_j^2/2 - (5p_j - 12.5) = \frac{1}{2}(p_j - 5)^2 > 0$). High: increasing in p_i, so $p_i = 10$, giving $10(10 - p_j) - \frac{1}{2}(10 - p_j)^2 = \frac{100 - p_j^2}{2}$. High $\geq$ undercut iff $p_j \leq 5\sqrt{2}$:

$$\phi(p_j) = 10 \text{ for } p_j \leq 5\sqrt{2}; \qquad \phi(p_j) = p_j^- \text{ (just below } p_j\text{, not attained) for } p_j > 5\sqrt{2}.$$

2. *Make a formal argument that symmetric prices cannot be a pure strategy Nash equilibrium.*

$p_1 = p_2 = p \leq 5$: earn $p^2/2 \leq 12.5$; deviate to 10: $\frac{100 - p^2}{2} \geq 37.5$. $p > 5$: earn $5p - 12.5$; deviate to $p - \varepsilon$: $\to p^2/2 > 5p - 12.5$. Profitable deviation either way.

3. *Make a formal argument that asymmetric prices cannot be a pure strategy Nash equilibrium.*

$p_j < p_i$: j earns $p_j^2/2$; any $p_j' \in (p_j, p_i)$ still undercuts and earns $p_j'^2/2 > p_j^2/2$. Profitable deviation.

4. *Use your result from 1 to find the bound of best response prices.*

Best responses lie in $[5\sqrt{2},\, 10] \approx [7.07, 10]$; upper bound $\bar{p} = 10$.

5. *Solve for the mixed strategy equilibrium using pricing bounds using the answer to 1.*

Symmetric CDF F on $[\underline{p}, 10]$, no atoms (tie earns $5p - 12.5 < p^2/2$, so mass is undercut). At $\underline{p}$ always lowest: $k = \underline{p}^2/2$. At 10 always highest: $k = E[(100 - X^2)/2]$. So $E[X^2] = 100 - \underline{p}^2$; $\underline{p} = 5\sqrt{2}$ would force an atom, so $\underline{p} < 5\sqrt{2}$.

6. *Solve for the Nash Equilibrium Mixed strategies.*

Indifference on $[\underline{p}, 10]$ (high firm sells $10-x$): $\int_{\underline{p}}^{p}[p(10-x) - \frac{1}{2}(10-x)^2]dF(x) + (1-F(p))\frac{p^2}{2} = k$. Differentiating twice, $S = 1 - F$, $t = p - 5$: $2t^2S'' + 6tS' + S = 0$, roots $r = -1 \pm \frac{1}{\sqrt{2}}$. With $F(10) = 1$:

$$F(p) = 1 - C\left[5^{\sqrt{2}}(p-5)^{-1-1/\sqrt{2}} - (p-5)^{-1+1/\sqrt{2}}\right], \quad p \in [\underline{p}, 10],$$

with $F(\underline{p}) = 0$ and $\underline{p} = 2(\underline{p} - 5)^2 f(\underline{p})$ giving $\underline{p} \approx 6.60$, $C \approx 0.287$.

7. *Calculate the Nash equilibrium expected profits.*

$E[\pi_i] = \underline{p}^2/2 \approx 21.8$ per firm.